Lec 3) Clustring and K-means

Q1) Clustering belongs to which type of machine learning?

B) unsupervised learning

Q2) A (age 25, income 60000), B (age 60, income 60600) why is the raw Euclidean distance misleading?

B) income has much larger scale and dominates the distance

Q3) which quantity does K-means minimize?

C) sum of squred distance from points to their cluster centroids.

Q4) In the update step of Lloyd's algorithm each centroid is moved to:-

B) The mean of the points assigned to it

Q5) Why does K-mans always termine?

B) J never increase and the number of partitions is finite

Q6) The elbow in the inertia plot suggests

A) The K after which extra clusters give small improvements

Q7) a(i) = 2 , b(i) = 5, its silhouette value is:-

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} = \frac{5-2}{\max(2,5)} = \frac{3}{5} = 0.60$$

B), 60

Q8) What problem does K-means ++ mainly address?

C) Poor, random initial centroids that cause bad local minima